Industrial Internship Report on

”CareerBoost AI Resume Analyzer”

Prepared by

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Executive Summary
This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT). This internship was focused on a project/problem statement provided by UCT. The project had to be completed along with the report within 6 weeks’ time. My project was CareerBoost AI Resume Analyzer — an AI-powered web application that evaluates resumes against ATS (Applicant Tracking System) standards, identifies missing skills, analyzes profile strength, and provides interview readiness recommendations to students and job seekers. This internship gave me a very good opportunity to get exposure to industrial/real-world problems and to design and implement an AI-based solution for it. It was an overall great experience to have this internship.

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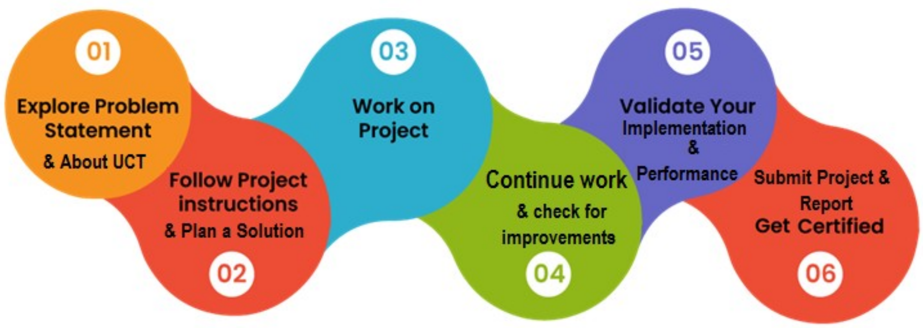
1 Preface

The six-week industrial internship provided an opportunity to understand real-world software development processes and apply technical knowledge to solve practical problems.

Resumes play a critical role in securing interviews and job offers, yet many strong candidates are filtered out by Applicant Tracking Systems (ATS) due to poor formatting, missing keywords, or unclear skill presentation. This made resume optimization a relevant and high-impact area for an internship project.

During this internship, I worked on developing CareerBoost AI Resume Analyzer using Python, Streamlit, and Gemini AI. The opportunity was given by upskill Campus (USC) and UniConverge Technologies Pvt Ltd (UCT), who structured the program so that the project and its report were to be completed within six weeks.

How the program was planned:



This internship improved my Python programming, AI integration, and prompt engineering skills, and gave me an overall understanding of how AI-driven career assistance products are designed, built, and deployed.

I would like to thank upskill Campus, The IoT Academy, UniConverge Technologies Pvt Ltd, and my mentors for their continuous guidance and support throughout this internship.

To my juniors and peers: take the problem statement seriously from day one, plan your six weeks in advance, and don’t hesitate to iterate on your solution — the AI integration and testing phases will need more time than you expect.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in the Digital Transformation domain, providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

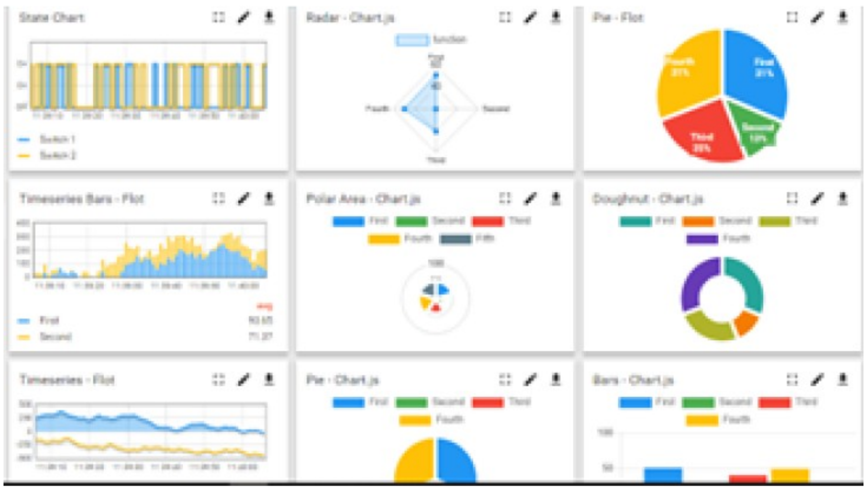
i. UCT IoT Platform (UCT Insight)

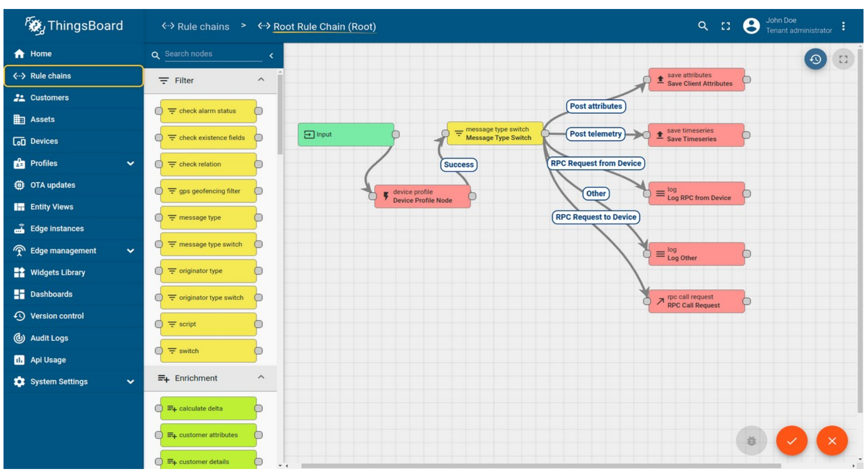
UCT Insight is an IOT platform designed for quick deployment of IOT applications, at the same time providing valuable "insight" for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSQL Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to:

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application (Power BI, SAP, ERP)
- Rule Engine





ii. Smart Factory Platform (Factory Watch)

FACTORY WATCH

Factory watch is a platform for smart factory needs.

It provides Users/ Factory:

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleash the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they want to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
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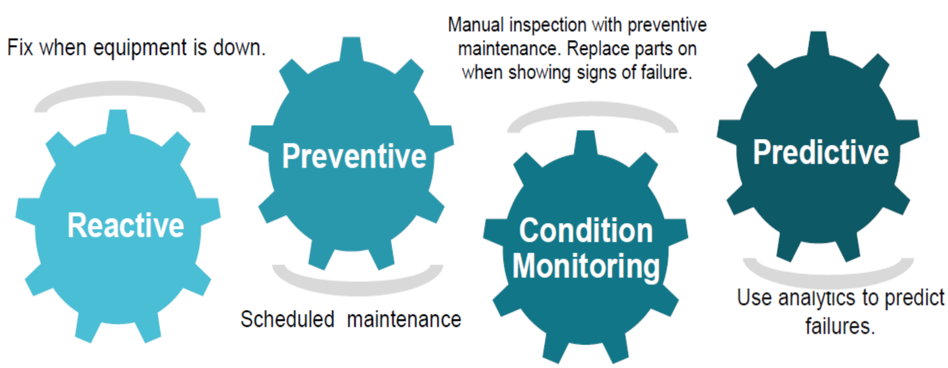
iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRaWAN technology and provides solutions in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

UCT is providing Industrial Machine health monitoring and Predictive maintenance solutions leveraging Embedded systems, Industrial IoT and Machine Learning Technologies by finding the Remaining Useful Life of various Machines used in the production process.

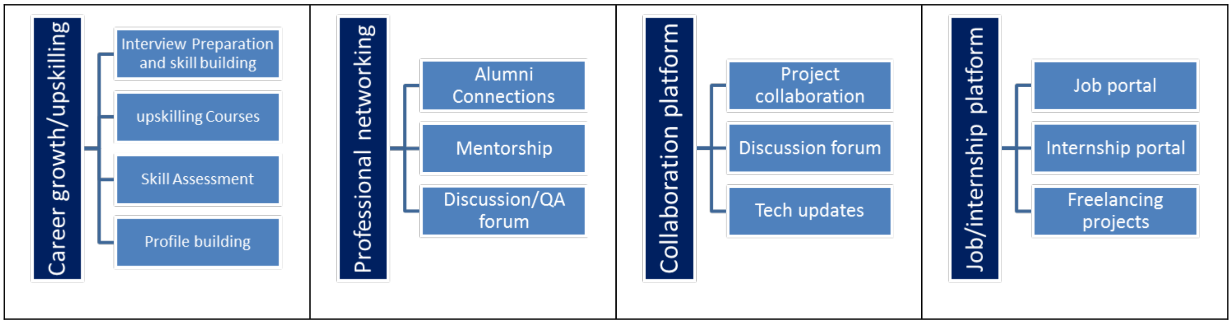


2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge Technologies has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.

Seeing the need for upskilling in a self-paced manner along with additional support services e.g. Internship, projects, interaction with industry, upskill Campus aims to upskill 1 million learners in the next 5 years.



2.3 The IoT Academy

The IoT academy is the EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to:

- get practical experience of working in the AI/software industry.
- solve a real-world problem faced by students and job seekers.
- have improved job prospects through a strong, AI-validated project portfolio.
- have improved understanding of AI integration, prompt engineering, and resume analytics.
- have personal growth like better communication and problem solving.

2.5 Reference

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[2] Streamlit Documentation, Available at: https://docs.streamlit.io/

[3] Google AI, "Gemini AI Documentation", Available at: https://ai.google.dev/

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[7] Streamlit Cloud Documentation, Available at: https://streamlit.io/cloud

[8] UniConverge Technologies Pvt Ltd and upskill Campus Internship Program Materials.

2.6 Glossary

Terms	Acronym
AI	Artificial Intelligence
ATS	Applicant Tracking System
PDF	Portable Document Format
UI	User Interface
API	Application Programming Interface
NLP	Natural Language Processing
Git	Version Control System
Dashboard	Visual Analytics Interface

3 Problem Statement

In the assigned problem statement:

Students and job seekers often struggle to create resumes that successfully pass Applicant Tracking Systems (ATS) before a human recruiter ever sees them. Across campus placement drives and off-campus applications alike, a large share of resumes are filtered out early — not because the candidate lacks skills, but because of:

- Poor or inconsistent resume formatting that ATS parsers cannot read correctly.
- Missing role-specific keywords that the ATS is configured to search for.
- Lack of clearly highlighted technical and professional skills.
- Weak or generic profile presentation that does not stand out.
- Insufficient preparation for the interview round that typically follows shortlisting.

Traditionally, the only way to identify and fix these issues is through manual review by a mentor, senior, or recruiter — a process that is slow, inconsistent, and not available to everyone, especially students from smaller colleges without strong placement cells.

The problem, therefore, is to design and develop an intelligent, AI-powered resume evaluation system that can automatically analyze a resume, calculate an ATS compatibility score, identify missing skills relevant to the candidate's target role, and generate clear, actionable recommendations — making this kind of feedback fast, consistent, and accessible to every student, not just those with access to expert mentors.

4 Existing and Proposed solution

Existing Solution

Today, resume evaluation is largely handled through manual review by mentors, seniors, or recruiters, or through generic online ATS checkers. These approaches have clear limitations:

- Time-consuming — manual review does not scale to hundreds of resumes.
- Subjective — feedback quality depends entirely on the reviewer's experience.
- Limited scalability — not accessible to every student at the same time.
- No automated, structured feedback loop for repeated improvement.

Proposed Solution

CareerBoost AI Resume Analyzer addresses these gaps with a complete AI-powered automation solution built using Python, Streamlit, and Gemini AI. Key value additions include:

- Automated AI-powered resume analysis using Gemini AI.
- Instant ATS score calculation based on formatting and keyword alignment.
- Automatic skill gap identification against role-relevant technical skills.
- Interview readiness assessment based on overall profile strength.
- Actionable, automated recommendations for resume improvement.
- Fast, scalable evaluation available to any user via cloud deployment — no manual reviewer required.

Together, these features save reviewer time, improve resume quality, enhance job readiness, and support long-term career growth for students and job seekers.

4.1 Code submission (Github link)

https://github.com/Praveen-6163/careerboost-ai-resume-analyzer

4.2 Report submission (Github link)

https://github.com/Praveen-6163/careerboost-ai-resume-analyzer

4.3 Live Project Deployment

https://careerboost-ai-resume-analyzer-yvvikannwf3emtebzaavzo.streamlit.app/

5 Proposed Design/ Model

The design flow below covers the start, intermediate processing stages, and final outcome of the CareerBoost AI Resume Analyzer system.

5.1 High Level Diagram

User → Resume Upload Module → PDF Parsing Engine → AI Processing Layer (Gemini AI) → ATS Scoring Engine → Skill Alignment Module → Interview Readiness Analyzer → Dashboard Visualization Layer

Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.2 Low Level Diagram

Step 1: User uploads resume in PDF format via the Streamlit interface.

Step 2: PDF content is extracted using PyPDF2 and Regex-based parsing.

Step 3: Gemini AI analyzes profile details, experience, and keyword density.

Step 4: ATS score is generated based on formatting and keyword alignment.

Step 5: Missing technical and professional skills are identified.

Step 6: Interview readiness is evaluated based on overall profile strength.

Step 7: Results are displayed through an interactive dashboard.

Figure 2: LOW LEVEL DIAGRAM OF THE SYSTEM

5.3 Interfaces

The interface and data flow for the CareerBoost AI Resume Analyzer pipeline is as follows:

- User uploads a resume using the Streamlit-based GUI (file upload widget).
- PDF Parsing Engine (PyPDF2 + Regex) extracts raw text and structural data from the resume.
- The AI Processing Layer (Gemini AI) receives the extracted text via a prompt-engineered API call.
- The application validates the AI response and computes the ATS score.
- Missing skills and improvement areas are identified and structured into a results object.
- Results, including ATS score, skill gaps, and interview readiness, are rendered on the Streamlit dashboard for the user.

Data flows in one direction through the pipeline (upload → parse → analyze → score → display), with no persistent server-side storage of resume content beyond the active session, which keeps the design simple and protects user privacy.

6 Performance Test

This is a very important part and defines why this work is meant for real industries, instead of being just an academic project.

Constraints identified for this project: AI response latency (Gemini API round-trip time), accuracy of ATS scoring against real-world ATS behaviour, and reliability of PDF text extraction across different resume formats and templates.

How these constraints were handled in the design: the PDF parsing module was built to handle multiple common resume layouts; prompt engineering was iterated to make Gemini AI's scoring more consistent; and the Streamlit interface shows a loading state so latency does not appear as a failure to the user.

6.1 Test Plan/ Test Cases

Test Case	Expected Result
Application startup	Streamlit app launches successfully
Resume upload (PDF)	File parsed and text extracted successfully
ATS score calculation	Accurate score generated based on resume content
Skill gap detection	Missing skills identified correctly for the target role
Interview readiness check	Relevant readiness insights generated
Corrupt/unsupported file upload	System shows a clear error message, no crash
AI service unavailable	System displays a graceful fallback/error message

6.2 Test Procedure

- Launch the application on Streamlit Cloud.
- Upload a sample resume in PDF format.
- Verify text extraction and AI analysis output.
- Check ATS score generation and skill gap output for correctness.
- Repeat with multiple resume formats/templates to test robustness.
- Monitor the dashboard for correct, complete display of results.

6.3 Performance Outcome

The system successfully performs resume analysis, ATS scoring, and skill gap detection with reliable accuracy across the resume formats tested. The AI processing layer responds within a few seconds per resume, and the dashboard displays results clearly without requiring any manual intervention. The application has been successfully deployed on Streamlit Cloud and integrated with GitHub for version control, confirming it is usable outside of a local development environment.

7 My learnings

This internship gave me practical, end-to-end exposure to building an AI-powered product — not just writing code, but also handling real user input, deploying to the cloud, and thinking about reliability.

Key learnings include:

- Building AI-powered Python applications with Streamlit as the front end.
- Integrating Gemini AI into an application and engineering prompts for consistent output.
- Extracting and cleaning real-world PDF resume content using PyPDF2 and Regex.
- Designing ATS scoring logic and skill-gap detection from scratch.
- Testing an AI-based system against varied, unpredictable real-world inputs.
- Deploying and maintaining an application using Streamlit Cloud and GitHub.

Overall, this internship strengthened both my technical and problem-solving skills, and gave me confidence to build and ship AI-based applications independently — experience that will directly support my career growth in software and AI development.

8 Future work scope

Due to the six-week time limitation, several enhancements could not be implemented in this cycle but are clearly identified for future work:

- Multi-language resume analysis, to support non-English resumes.
- Resume builder integration, so users can fix issues directly within the app.
- LinkedIn profile analysis alongside the resume for a fuller picture.
- AI-generated cover letters tailored to each job description.
- A real-time job recommendation system based on the analyzed profile.
- Advanced, role-specific career guidance features.

These additions would extend CareerBoost AI Resume Analyzer from a resume-scoring tool into a more complete, end-to-end AI career assistant.